

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Inferring Probability Models

Chapter 9 — Maximum Likelihood and Bayesian Inference

Dr. Khalil Chebil

Probability & Statistics for Computer Science

Chapter Outline | What we will cover

Maximum Likelihood Estimation
Properties of MLEs
Bayesian Inference: The Big Idea
Credible Intervals & Sequential Updating

Conjugate Priors
MAP Estimation & the Normal Model
Applications
Summary

Learning Objectives

- ▶ Derive and compute the **maximum likelihood estimator**
- ▶ Apply **Bayes' theorem** to update beliefs about parameters
- ▶ Use **conjugate priors** for closed-form posteriors
- ▶ Compare **MAP** and MLE, and see Bayesian methods in practice

Section 1

Maximum Likelihood Estimation

The Core Idea | Pick the parameter that makes the data most probable

The likelihood function

For iid data $x_1, \dots, x_n$:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

Log-likelihood

Sums are easier than products:

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$$

The MLE

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$$

MLE for the Bernoulli | The sample proportion, derived from first principles

$$\ell(p) = \sum_i [x_i \log p + (1 - x_i) \log(1 - p)].$$

Setting $\ell'(p) = 0$:

Result

$$\hat{p}_{\text{MLE}} = \frac{1}{n} \sum_i x_i = \bar{x}$$

The MLE for a coin's bias is simply the observed fraction of heads.

MLE for the Normal and the Poisson | The same recipe, two more distributions

Model	MLE
Normal (μ, σ^2)	$\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$
Poisson (λ)	$\hat{\lambda} = \bar{x}$

Pattern

In each case the MLE is a familiar sample statistic — MLE *justifies* using the sample mean/proportion, it doesn't just assume it.

Section 2

Properties of MLEs

Four Properties Worth Knowing | Why statisticians trust the MLE

Property	Meaning
Consistency	$\hat{\theta} \rightarrow \theta$ as $n \rightarrow \infty$
Asymptotic normality	$\hat{\theta}$ is approximately normal for large n
Efficiency	Achieves the lowest possible variance asymptotically
Invariance	MLE of $g(\theta)$ is $g(\hat{\theta})$, for any function g

Confidence intervals from the MLE

Asymptotic normality gives $\hat{\theta} \pm z^* / \sqrt{I(\hat{\theta})}$, using the Fisher information $I(\theta)$.

Section 3

Bayesian Inference: The Big Idea

The Bayesian Philosophy | Treat the parameter itself as uncertain

MLE's view

θ is a fixed, unknown constant — we produce a single best guess.

Bayesian view

θ has a **probability distribution** reflecting our uncertainty, updated as data arrives.

Bayes' theorem for parameters

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) p(\theta)}{p(\text{data})} \propto \underbrace{p(\text{data} \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

Worked Example: Coin Flipping | Beta prior meets Bernoulli data

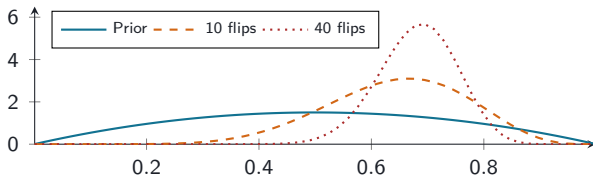
Prior: $p \sim \text{Beta}(\alpha, \beta)$. Observe k heads in n flips.

$$\text{posterior} \propto p^k (1 - p)^{n-k} \cdot p^{\alpha-1} (1 - p)^{\beta-1}$$

Result

$$p \mid \text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

Still a Beta — the posterior stays in the same family as the prior.



Section 4

Credible Intervals & Sequential Updating

Credible Intervals | A direct probability statement

Definition

A range $[a, b]$ such that

$$P(a \leq \theta \leq b \mid \text{data}) = 0.95$$

vs. confidence intervals

A credible interval *does* mean "95% probability θ is in here" — the interpretation frequentist CIs lack.

Sequential Updating | Today's posterior is tomorrow's prior

Bayesian learning

After observing batch 1, the posterior becomes the prior for batch 2 — beliefs update smoothly as evidence accumulates.

Posterior predictive

Probability the *next* flip is heads, averaging over our uncertainty in p :

$$P(\text{next} = H \mid \text{data}) = \frac{\alpha + k}{\alpha + \beta + n}$$

5

Section 5

Conjugate Priors

Conjugate Priors | When the posterior stays in the prior's family

Likelihood	Conjugate prior	Posterior
Binomial (n, p)	$\text{Beta}(\alpha, \beta)$	$\text{Beta}(\alpha + k, \beta + n - k)$
Poisson (λ)	$\text{Gamma}(\alpha, \beta)$	$\text{Gamma}(\alpha + \sum x_i, \beta + n)$
Normal (known σ^2)	$\text{Normal}(\mu_0, \tau_0^2)$	Normal (precision-weighted)

Why it matters

Conjugacy gives a **closed-form** posterior — no numerical integration required.

Normal–Normal: A Precision-Weighted Average | Combining prior belief with data

Prior $N(\mu_0, \tau_0^2)$, data mean $\bar{x}$ from n points with known σ^2 :

Posterior mean

$$\mu_n = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

Interpretation

A weighted average of the prior mean and the data mean, weighted by their respective **precisions** (1/variance) — more data pulls the estimate toward $\bar{x}$.

Section 6

MAP Estimation & the Normal Model

Maximum A Posteriori (MAP) Estimation | A single best guess, informed by the prior

Definition

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \text{data}) = \arg \max_{\theta} L(\theta) p(\theta)$$

MAP as regularized MLE

A Gaussian prior on parameters $\Rightarrow$ MAP = ridge regression; a Laplace prior $\Rightarrow$ MAP = LASSO.

MAP vs. MLE vs. posterior mean

As $n \rightarrow \infty$, the prior's influence vanishes and MAP converges to the MLE. For a symmetric, unimodal posterior, MAP equals the posterior mean.

Bayesian Inference for the Normal | Why Student's t appears

Both μ and σ^2 unknown

The conjugate prior is **Normal-Gamma**; the marginal posterior for μ integrates out the uncertainty in σ^2 .

The result

That marginalization produces a **Student's t -distribution** for μ — the same distribution that shows up in Chapter 6's confidence intervals.



Section 7

Applications

Bayesian A/B Testing | A more intuitive answer than a p -value

Frequentist framing

"Reject $H_0 : p_A = p_B$ at $\alpha = 0.05$ " — a yes/no verdict.

Bayesian framing

$P(p_B > p_A \mid \text{data})$ — a direct probability that variant B is better, easy to explain to stakeholders.

More Applications | The same Beta-Binomial machinery, everywhere

Application	What's being estimated
Quality control	True defect rate, updated batch by batch
Email spam filtering	Probability a message is spam given word features
Sequential coin testing	Bias of a coin, refined flip by flip
ML hyperparameter tuning	Best configuration via Bayesian optimization

8

Section 8 Summary

MLE vs. Bayesian: When to Use What | Two complementary toolkits

Approach Strengths

MLE Simple, objective, works well with plenty of data

Bayesian Incorporates prior knowledge, natural with small data, gives full distributions, direct probability statements

Chapter Summary | Key takeaways

Estimate

MLE finds the parameter that best explains the data; Bayesian inference produces a full posterior distribution.

Update

Conjugate priors (Beta-Binomial, Gamma-Poisson, Normal-Normal) give closed-form posteriors.

Decide

MAP blends prior and likelihood into one point estimate — a bridge between the two philosophies.

The journey

From describing data (Part I) through probability (Part II) to inference (Part III) — the full toolkit of applied statistics.

Thank you!

Questions & Discussion

End of Course — Good luck!